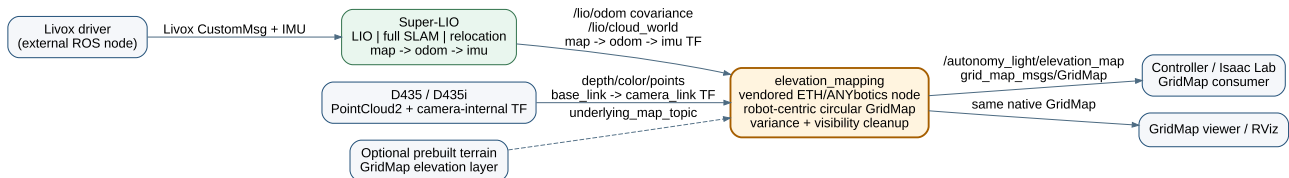


autonomy_light software architecture

Super-LIO + ETH/ANYbotics robot-centric elevation mapping · 2026-08-04

Design decision. This repository no longer contains an elevation-map algorithm, custom message, DDS bridge, or D435 merge node. It brings up one vendored ETH/ANYbotics elevation mapping node; Super-LIO remains the single owner of LIO, full SLAM, relocation, and `map → odom`.



Node responsibilities

Node	Owner	Responsibility
Livox driver	external	Publishes one Livox CustomMsg and IMU stream.
Super-LIO	external	LiDAR–IMU odometry, ESKF covariance, full SLAM, saved-map relocation, and <code>map → odom → imu</code> .
D435/D435i driver	external	Publishes one PointCloud2 stream with its calibrated TF chain.
<code>elevation_mapping</code>	vendored ETH/ANYbotics	Robot-centric circular grid, sensor-specific variance, pose-covariance prediction, Bayesian fusion, and visibility cleanup.
<code>autonomy_light</code>	this repository	Configuration, process lifecycle, and calibrated static TF only.

Data and TF contract

Direction	Topic / frame	Meaning
input	<code>/lio/odom</code>	Super-LIO pose and 6×6 covariance for motion uncertainty propagation.
input	<code>/lio/cloud_world</code>	Registered LiDAR cloud, transformed to the map at its timestamp.
optional input	<code>/camera/.../depth/color/points</code>	D435/D435i cloud; directly fused with the structured-light model.
output	<code>/autonomy_light/elevation_map</code>	Native <code>grid_map_msgs/msg/GridMap</code> , layers: elevation, upper/lower bound,

		uncertainty range.
TF	map → odom → imu → base_link → lidar_link base_link → camera_link → camera_depth_optical_frame	Super-LIO owns dynamic transforms; launcher owns the calibrated static mounts, while the D435 driver owns camera-internal TF.

The local grid follows `base_link` in `map` with Grid Map's circular buffer. It preserves overlapping posterior cells while moving and clears only newly exposed strips. Unobserved terrain remains `NaN`; no arbitrary startup floor value is inserted.

Uncertainty and cleanup

Each input uses its own laser or D435 noise model. At each cloud timestamp the elevation node obtains the Super-LIO covariance, predicts map uncertainty due to robot motion, and applies the measurement update in the map frame. Its visibility cleanup ray-traces newer observations to remove penetrated stale cells, addressing lingering wall and overhanging-obstacle artifacts without an additional local implementation.

Prebuilt terrain input

`config/elevation_mapping.yaml` owns mapper inputs, local grid, fusion and cleanup. `config/autonomy_light.yaml` contains the one canonical map/base TF names, calibration and Super-LIO/Livox inputs; the launcher binds those shared frame names to the mapper. Use `--elevation-config FILE` to select another mapper profile; the output topic is a launcher remap (`AUTONOMY_LIGHT_ELEVATION_MAP_TOPIC`).

`underlying_map_topic` accepts a prebuilt `GridMap` carrying an `elevation` layer. A Super-LIO PCD remains the map used by `--map` relocation and is not silently reinterpreted as terrain. This keeps localization and terrain-map contracts explicit.

Operational commands

- `./build.sh` installs/builds Livox, Super-LIO, Grid Map, vendored KINDR/KINDR ROS, the mapper, and bringup.
- `./launch.sh --vis` opens a latest-frame GridMap elevation viewer; `--rviz` opens the GridMap RViz display.
- `./mapping.sh --output maps/site.pcd` runs Super-LIO full SLAM and writes the optimized PCD at shutdown.